PEMEtrexed and CISplatin Therapy

Please refer to NCCP Regimen 00849 Nivolumab 360mg and Chemotherapy for relevant information when used in combination with nivolumab

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Treatment of chemotherapy naïve patients with unresectable malignant pleural mesothelioma.	C45	00317a	N/A
First line treatment of patients with locally advanced or metastatic non-small cell lung cancer (NSCLC).	C34	00317b	N/A
In combination with nivolumab for the neoadjuvant treatment of resectable NSCLC at high risk of recurrence in adult patients whose tumours have PD-L1 expression ≥ 1% **(3 cycles only)** **(This combination is available in NCIS (00849.5))**	C34	00317c	Nivolumab: ODMS 01/05/2024 Chemotherapy: N/A

*This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

For the treatment of malignant pleural mesothelioma PEMEtrexed and CISplatin are administered once every 21 days for a maximum of 4-6 cycles unless the patient experiences disease progression or unacceptable toxicity.

For the first line treatment of locally advanced or metastatic NSCLC PEMEtrexed and CISplatin are administered once every 21 days for a maximum of 4-6 cycles followed by maintenance PEMEtrexed (Reference NCCP regimen 00222) unless the patient experiences disease progression or unacceptable toxicity.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

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Admin. Order	Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	1	PEMEtrexed	500mg/m^2	IV infusion	100mL NaCl 0.9% over 10 minutes	Every 21 days
2	1	[a]CISplatin	75mg/ m^2	IV infusion	1000mL NaCl 0.9% over 60 minutes to start 30 minutes after completion of PEMEtrexed	Every 21 days
		Folic Acid or multivitamin containing 350-1000 micrograms folic acid	350-1000 micrograms[b]	PO		

[a] **Pre and post hydration therapy required for CISplatin**
See local hospital policy recommendations.
Suggested prehydration for CISplatin therapy:

- Administer 1000mL NaCl 0.9% over 60 minutes
- Administer CISplatin as described above

Post hydration:

- Administer 10mmol magnesium sulphate ($MgSO_4$) and 20mmol potassium chloride (KCl) in1000mL NaCl 0.9% over 2 hours (Refer to relevant local hospital policy for advice on administration of electrolyte infusions).
- Mannitol 10% may be used as per local policy to induce diuresis, although there is no conclusive evidence that this is required. The routine use of furosemide to increase urine flow is not recommended unless there is evidence of fluid overload.

[b] At least five doses of folic acid must be taken during the seven days preceding the first dose of PEMEtrexed, and dosing must continue during the full course of therapy and for 21 days after the last dose of PEMEtrexed.
See Premedications for further treatment required.

PEMEtrexed is physically incompatible with diluents containing calcium, including lactated Ringer's injection and Ringer's injection.

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

ELIGIBILITY:

- Indications as above
- ECOG 0-2

EXCLUSIONS:

- Hypersensitivity to PEMEtrexed, CISplatin or any of the excipients
- PEMEtrexed
  - Creatinine clearance < 45mL/min
- CISplatin
  - Pre-existing renal impairment
  - Pre-existing neuropathies ≥ grade 2
  - Significant hearing impairment/tinnitus
- Pregnancy
- Breastfeeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Audiology referral as clinically indicated

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Regular tests:

- FBC, renal and liver profile prior to each cycle

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant

Haematological:

- Treatment may be delayed to allow sufficient time for recovery
- Upon recovery patient should be retreated using the guidelines below

Table 1: Dose modifications for haematological toxicity of PEMEtrexed and CISplatin

ANC x 10^9 /L	Comment	Platelets x 10^9 /L	Comment
≥ 1.5	100%	≥100	Continue
1.0 to < 1.5	Consult with team, if clinically well continue treatment	75 - <100	Consult with team, if clinically well consider continuation
0.5 to < 1.0	Delay until treatment recovery	50 - < 75	Delay until treatment recovery
< 0.5	Delay until treatment recovery and reduce PEMEtrexed and CISplatin by 25%	<50	Delay until treatment recovery and reduce PEMEtrexed and CISplatin by 25%
Febrile neutropenia	Delay until treatment recovery and reduce PEMEtrexed and CISplatin by 25%	<50 with bleeding	Delay until treatment recovery and reduce PEMEtrexed and CISplatin by 50%

Renal and Hepatic Impairment:

Table 2: Dose modification of CISplatin and PEMEtrexed in renal and hepatic impairment

	Renal Impairment		Hepatic Impairment	
CISplatin	**CrCl (mL/min)**	**Dose**	No need for dose adjustment is expected.	
	50-59	75% of the original dose		
	40-49	50% of the original dose		
	<40	Not recommended		
	Haemodialysis	50% of the original dose may be considered		
PEMEtrexed	**CrCl (mL/min)**	**Dose**		**Dose**
	≥45	No dose adjustment is needed	Mild and moderate	No need for dose adjustment is expected
	<45	Not recommended	Severe	Not recommended, based on the risk of PEMEtrexed induced liver dysfunction.
	Haemodialysis	Not recommended		
Renal and hepatic recommended as per Giraud et al 2023.				

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Management of adverse events:

Table 3: Dose Modification of PEMEtrexed and CISplatin for Adverse Events

Adverse effect	Recommended dose modification
Diarrhoea Grade ≥ 3 and any diarrhoea irrespective of grade requiring hospitalisation	Withhold treatment until resolution to less than or equal to patient's pre-therapy value. Resume at 75% of previous dose for both PEMEtrexed and CISplatin.
Mucositis Grade ≥3	Withhold treatment until resolution to less than or equal to patient's pre-therapy value. Resume at 50% of previous dose for PEMEtrexed and at 100% of previous dose for CISplatin.
Neurotoxicity Grade 2 Grade ≥3 or any recurrence of neurotoxicity	Withhold treatment until resolution to less than or equal to patient's pre-therapy value. Resume at 100% dose for PEMEtrexed and reduce CISplatin by 50%. Discontinue
Other Grade ≥3	Withhold treatment until resolution to less than or equal to patient's pre-therapy value. Resume at resume at 75% of previous dose PEMEtrexed and CISplatin
Grade ≥3 toxicity after 2 dose reductions	Consider discontinuation of both drugs. Discuss with consultant.
Allergic reactions: Grade 3 ≥	Clinician decision. Consider discontinuation of whichever drug caused the allergy.

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

CISplatin: High **(Refer to local policy)**
PEMEtrexed: Low **(Refer to local policy)**

For information:
Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists. Information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for Inclusion in NCIS (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for Inclusion in NCIS (Haemato-oncology) - Available on the NCCP website

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PREMEDICATIONS:

- A corticosteroid should be given the day prior to, on the day of, and the day after PEMEtrexed administration. The corticosteroid should be equivalent to 4 mg of dexAMETHasone administered orally twice a day
- Intramuscular injection of vitamin B_{12} (hydroxycobolamin) (1,000 micrograms) in the week preceding the first dose of PEMEtrexed and once every three cycles thereafter. Subsequent vitamin B_{12} injections may be given on the same day as PEMEtrexed

OTHER SUPPORTIVE CARE:

- Hydration pre and post CISplatin administration (Reference local policy or see recommendations above)

ADVERSE EFFECTS

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

REFERENCES:

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Version	Date	Amendment	Approved By
1			Prof Maccon Keane
2	02/05/2018	Updated with new Regimen Template, updated exclusions, CISplatin hydration recommendations, dosing in renal impairment and adverse reactions	Prof Maccon Keane
3	29/04/2020	Reviewed. Update of adverse events.	Prof Maccon Keane
4	23/9/2020	Clarification of number of cycles	Prof Maccon Keane
5	24/06/2021	Updated CISplatin hydration protocol	Prof Maccon Keane
6	01/05/2024	New indication for nivolumab in the neoadjuvant setting and reference to relevant nivolumab regimen added. Amended CISplatin infusion time.	Prof Maccon Keane
7	17/10/2025	Regimen reviewed. Updated: - Treatment outline - Exclusion criteria - Baseline testing - Haematological dose modifications – nadir counts removed - Renal and hepatic dose modifications to align with Giraud et al 2023. - Dose modification recommendations for non-haematological toxicities. - Dose modifications recommendations in adverse events - Updated regimen in line with NCCP standardisation	Prof Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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